

Data Link Layer Architecture & Network Addressing

Transportation Analogies, IPv4 vs. MAC Addressing, ARP Mechanisms, Switch Self-Learning, and Solved Conceptual Exam Problems

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Abstract

This academic document provides an exhaustive, structured synthesis of the Data Link Layer (Layer 2) of the OSI model. It covers transportation analogies for data transmission, framing, and media access control. It offers a detailed technical contrast between Network Layer (IPv4) and Data Link Layer (MAC) addressing, including bit-level analysis of Organizationally Unique Identifiers (OUIs) and Unicast/Multicast/Local flag bits. Furthermore, it details the operational workflow of the Address Resolution Protocol (ARP) across both single LANs and routed multi-subnet boundaries. Finally, it analyzes switch forwarding algorithms, self-learning tables, and provides rigorous step-by-step solutions for all slide-based conceptual exam questions.

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1 Data Link Layer Fundamentals & Transportation Analogies

To understand how data traverses a complex internetwork, the physical and logical operations of the Data Link Layer (Layer 2) can be understood through transportation analogies.

1.1 The Transportation Analogy Paradigm

Real-World Transportation Analogy: Imagine a tourist traveling from **Dhaka, Bangladesh** to **Zurich, Switzerland**. The journey requires taking a taxi to the airport, an international flight to Frankfurt, a train to Zurich, and a bus to the final hotel.

- **The Traveler / Cargo:** Represents the **Datagram** (Network Layer Packet).
- **The Route / Highway / Rail Track:** Represents the **Communication Link** (copper cable, fiber optic, wireless spectrum).
- **Mode of Transportation (Taxi, Plane, Train):** Represents the **Link Layer Protocol** (Ethernet, Wi-Fi 802.11, PPP).
- **Travel Ticket / Encapsulated Pass:** Represents the **Frame** created at the Data Link Layer by adding a header and trailer to the datagram.

1.2 Core Services of the Link Layer

1. **Framing:** Encapsulates the Layer 3 IP datagram into a Layer 2 **Frame** by appending a *Header* (containing Source and Destination MAC addresses) and a *Trailer* (containing Frame Check Sequence / FCS for error detection).
2. **Link Access & Medium Access Control (MAC):** Governs the rules for how multiple devices share a single physical transmission medium (e.g., CSMA/CD in legacy Ethernet, CSMA/CA in Wi-Fi).
3. **Per-Hop Error Control:** Link-layer protocols detect (and sometimes correct) bit errors caused by physical noise over each individual link using Cyclic Redundancy Checks (CRC). This contrasts with the Transport Layer (TCP), which performs end-to-end error control.
4. **Flow Control:** Prevents a fast-transmitting sending node from overwhelming a slower receiving node on the same physical segment.

1.3 Duplex Transmission Modes & Physical Implementation

Where is the Link Layer Implemented? The Data Link Layer is implemented in the **Network Interface Controller (NIC)** (also known as the network adapter) present in every networked host and router. The NIC combines hardware (MAC chipsets, physical transceivers), firmware, and software drivers to directly interface with the system bus and physical media.

Table 1: Comparison of Duplex Transmission Modes

Dimension	Half-Duplex	Full-Duplex
Data Flow	Bidirectional, but only one direction at a time.	Simultaneous bidirectional transmission.
Real-World Analogy	Walkie-talkie radio / single-lane bridge.	Two-way highway / modern telephone call.
Network Device	Legacy Ethernet Hubs / Coaxial Bus.	Modern Ethernet Switches (Switched LANs).
Collision Risk	High (requires CSMA/CD).	Zero collisions on dedicated switch links.

2 Addressing Paradigms: IPv4 Addresses vs. MAC Addresses

Network communication requires two primary addressing tiers: logical network addresses (Layer 3) and physical hardware addresses (Layer 2).

2.1 Comparative Architectural Breakdown

Table 2: Comprehensive Structural Comparison: IP Address vs. MAC Address

Attribute	IP Address (IPv4)	MAC Address (Ethernet)
OSI Layer	Layer 3 (Network Layer)	Layer 2 (Data Link Layer)
Address Size	32 Bits (4 Bytes)	48 Bits (6 Bytes / Octets)
Representation	Dotted-decimal (e.g., 192.168.1.10)	Hexadecimal notation (e.g., 00:10:65:C0:00:01)
Structure	Hierarchical (Network ID + Host ID)	Flat (No topology-based prefixing)
Portability	Dynamic / Changeable when moving subnets	Permanent / Burned into NIC ROM
Primary Role	Global end-to-end packet routing	Local hop-to-hop frame delivery within LAN

2.2 Detailed Bit-Level MAC Address Structure

A 48-bit MAC address consists of 6 octets, typically formatted in hexadecimal. It is split into two equal 24-bit halves:

2.2.1 Crucial Flag Bits in the First Octet

When analyzing a MAC address in binary, two specific bits in the **first octet** determine its delivery type and administrative status:

1. The Least Significant Bit of the 1st Octet (b_0 / LSB): Unicast vs. Multicast Flag

- If $b_0 = 0$: The address is a **Unicast** address (destined for a single specific host).
- If $b_0 = 1$: The address is a **Multicast** address (destined for a group of hosts).

Assigned by IEEE to Manufacturer

Assigned by Vendor (Serial ID)

Organizationally Unique Identifier (OUI) First 24 Bits (3 Octets / 6 Hex Digits)	Network Interface Controller (NIC) Specific Last 24 Bits (3 Octets / 6 Hex Digits)
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Bit-Level Structure of 1st Octet ($b_7b_6b_5b_4b_3b_2b_1b_0$):

Figure 1: 48-Bit Ethernet MAC Address Organizational Structure.

- *Hexadecimal Quick-Check Rule:* Convert the 2nd hexadecimal digit of the address to binary. If the LSB is 1 (i.e., the second hex digit is odd: 1, 3, 5, 7, 9, B, D, F), the MAC address is Multicast!

2. The 2nd Least Significant Bit of the 1st Octet (b_1): Global vs. Local Flag

- If $b_1 = 0$: The address is **Globally Unique** (OUI enforced by IEEE).
- If $b_1 = 1$: The address is **Locally Administered** (overridden manually by software/administrator).

2.3 Types of MAC Addresses

- **Unicast MAC Address:** Points to a single specific NIC (e.g., 00:10:65:C0:00:01).
- **Multicast MAC Address:** Points to a multicast group. Ethernet multicast addresses always start with 01:00:5E for IPv4 multicast mapping.
- **Broadcast MAC Address:** Destination address where all 48 bits are set to 1 (FF-FF-FF-FF-FF-FF). Received and processed by every host on the Layer 2 local broadcast domain.

2.4 Why is a MAC Address Called a "Flat" Address?

Unlike IP addresses, which are **hierarchical** (where the network prefix identifies the subnet location, allowing routers to aggregate thousands of routes into a single prefix), MAC addresses are **flat**. The address contains no geographical or network-topological location data. Moving a computer from Dhaka to London changes its IP address, but its MAC address remains unchanged.

3 Address Resolution Protocol (ARP) & LAN Operations

Because Network Layer packets use logical IP addresses, but local Ethernet switches forward frames based strictly on Layer 2 MAC addresses, a translation mechanism is required: the ****Address Resolution Protocol (ARP)****.

3.1 ARP Workflow within a Single LAN

When Host A (192.168.1.10) wants to send a datagram to Host B (192.168.1.12) on the same local subnet, but does not know Host B's MAC address:

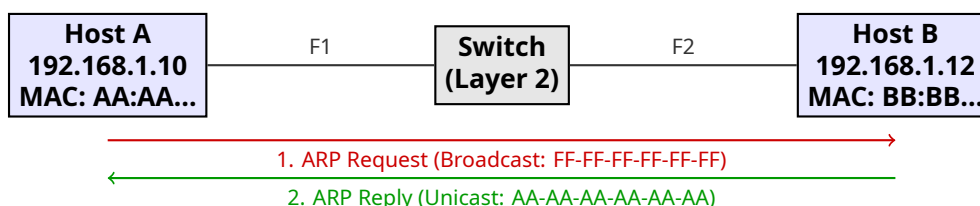


Figure 2: Single LAN ARP Resolution Sequence.

- 1. ARP Request (Broadcast):** Host A constructs an ARP Request frame asking: *"Who has IP 192.168.1.12? Tell Host A at 192.168.1.10."*
 - **Source MAC:** AA-AA-AA-AA-AA-AA (Host A)
 - **Destination MAC:** FF-FF-FF-FF-FF-FF (Broadcast)
- 2. Switch Flooding:** The Layer 2 switch receives the broadcast frame and floods it out all active ports except the incoming port.
- 3. Processing at Hosts:** All connected hosts receive the frame. Host B recognizes its own IP address in the target field, while all other hosts drop the packet.
- 4. ARP Reply (Unicast):** Host B builds a unicast ARP Reply directly back to Host A: *"192.168.1.12 is at MAC BB-BB-BB-BB-BB-BB."*
 - **Source MAC:** BB-BB-BB-BB-BB-BB (Host B)
 - **Destination MAC:** AA-AA-AA-AA-AA-AA (Host A)
- 5. Caching:** Host A receives the reply and stores the mapping in its local ****ARP Table****.

3.2 ARP Table Structure & TTL Expiration

Each host maintains a dynamic ARP cache in RAM:

Table 3: Sample Host ARP Cache Table

IP Address	MAC Address	Time-To-Live (TTL)
192.168.1.12	BB-BB-BB-BB-BB-BB	20 min
192.168.1.1	00-11-22-33-44-55	15 min

Why ARP Tables Have a TTL (Plug-and-Play Nature): ARP is completely **plug-and-play** (requires no manual user configuration). ARP table entries are assigned a ****Time-To-Live (TTL)**** (typically 10–20 minutes). If a host's Network Interface Controller (NIC) is replaced or its IP address changes, stale ARP entries expire automatically, preventing permanent communication failures due to MAC address mismatches.

3.3 Cross-Subnet ARP: Routing Across Multiple LANs

When Host A (192.168.1.10/24) wants to transmit a packet to Host O (10.0.0.5/24) on a remote network separated by a router:

Golden Rule of Cross-Subnet ARP: Routers strictly drop Layer 2 broadcast frames! An ARP broadcast request generated on Subnet A will **never** pass through a router to reach Subnet B.

3.3.1 Cross-Subnet Step-by-Step Resolution Workflow

1. Host A compares destination IP 10.0.0.5 with its own subnet mask and determines Host O is on a ****remote network****.
2. Host A realizes it must forward the datagram to its ****Default Gateway**** (the local interface of Router R1, e.g., 192.168.1.1).
3. Host A checks its ARP table for Default Gateway IP 192.168.1.1. If absent, Host A broadcasts an ARP Request for ****Default Gateway R1's MAC address**** (NOT Host O's MAC!).
4. Host A encapsulates the IP packet (Source IP: 192.168.1.10, Destination IP: 10.0.0.5) inside an Ethernet frame with ****Destination MAC = Router R1's local interface MAC****.
5. Router R1 receives the frame, strips the Layer 2 header, inspects the Layer 3 destination IP, consults its routing table, forwards the packet across WAN links, and eventually performs a ****new ARP request**** on the remote LAN using its outgoing interface MAC address.

4 Switching Architecture vs. Hubs & Routers

An Ethernet switch is an active Layer 2 networking device that selectively forwards frames based on hardware addresses.

4.1 Switches vs. Hubs vs. Routers

Table 4: Comprehensive Comparison: Hub vs. Switch vs. Router

Property	Hub	Switch	Router
OSI Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)
Forwarding Metric	Electrical signal repetition	Destination MAC Address	Destination IP Address
Forwarding Mechanism	Floods all frames to all ports	Selective forwarding (via Self-Learning Table)	Table lookup via routing algorithms (Dijkstra/DV)
Collision Domains	Single shared domain across all ports	Separate micro-segment per port	Separate domain per interface
Broadcast Domains	Single broadcast domain	Single broadcast domain	Blocks Layer 2 broadcasts
Duplex Capability	Half-Duplex only	Full-Duplex supported	Full-Duplex supported

The Switch Collision Limitation: While a switch provides full-duplex isolated collision domains on every port (allowing Host A and Host B to communicate simultaneously while Host C and Host D communicate), **a switch cannot transmit two frames simultaneously out of the exact same physical port or from the exact same source interface at the same microsecond**. If multiple frames are destined for the same egress port at once, the switch queues them in its buffer.

4.2 Switch Self-Learning Algorithm

A switch operates transparently (hosts are completely unaware of its presence). It builds its **Switch Table** (MAC-to-Interface Mapping Table) automatically using **Self-Learning**:

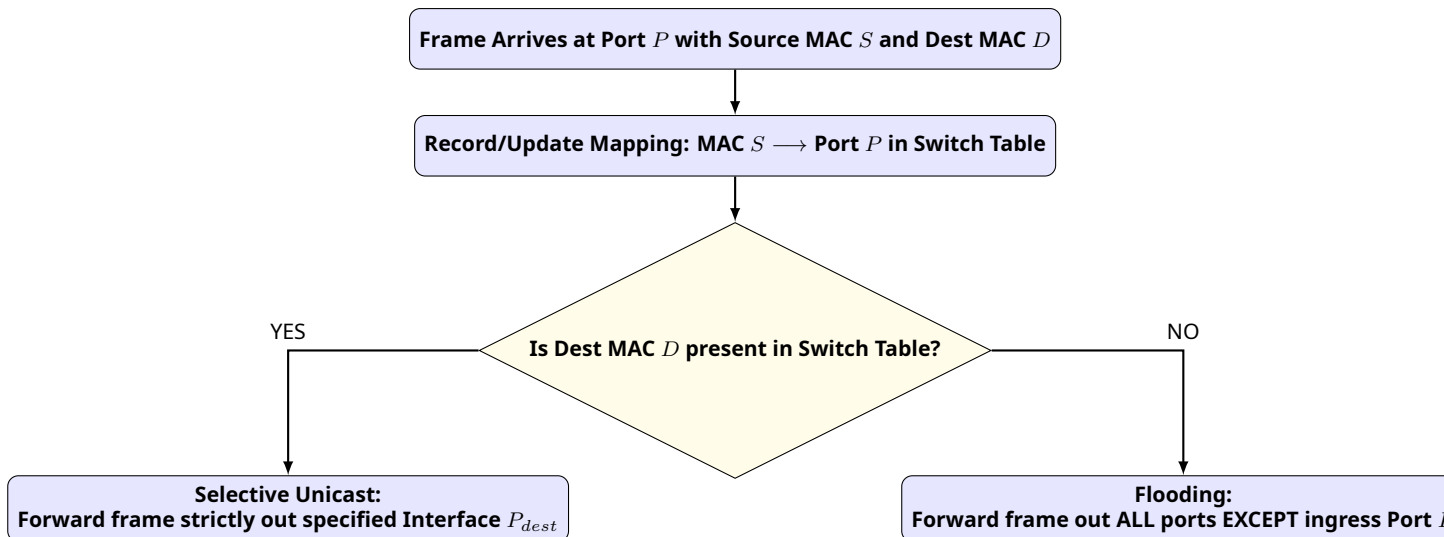


Figure 3: Switch Self-Learning and Frame Forwarding Logic Flowchart.

5 Solutions to Slide-Based Conceptual Exam Questions

Below are the complete, rigorous solutions to all exam problems presented in the course slides and attached images.

5.1 Exam Problem 1: MAC Address OUI & Flat Address Properties

Slide Exam Question 1 (Referencing Images 4 & 7): Given your MAC address is *AF:CC:FE:12:23:40*.

- I. Identify the OUI part of the MAC address.
- II. Discuss why the MAC address is considered to be a flat address.
- III. Specify the two parts of a MAC address. Identify which bit signifies if it's a unicast or a multicast address.

Step-by-Step Solution:

- I. **OUI Identification:** The Organizationally Unique Identifier (OUI) consists of the first 24 bits (3 octets / 6 hexadecimal digits) of the MAC address.

$$\text{OUI Part} = \mathbf{AF : CC : FE} \quad (1)$$

(The remaining portion 12:23:40 is the Vendor/NIC-specific serial number).

- II. **Why MAC is a "Flat" Address:** A MAC address is termed "flat" because it lacks any hierarchical network partitioning or location-based prefixing. Unlike a hierarchical IPv4 address (where the subnet portion changes based on geographic/network location to allow route aggregation), a MAC address is permanently bound to the hardware NIC ROM regardless of where the device is physically moved in the world.

- III. **Two Parts & Unicast/Multicast Bit Identification:**

- **Two Parts:** (1) The **OUI** (First 24 bits) and (2) The **NIC-Specific Identifier** (Last 24 bits).
- **Unicast/Multicast Bit:** It is the ****Least Significant Bit (LSB) of the first octet**** (often denoted as bit b_0 of octet 1).

- *Analysis for given MAC AF:CC:FE:12:23:40:* First octet = $AF_{16} = 1010\ 1111_2$. The LSB (b_0) is **1**, which signifies that this address is a **Multicast MAC Address**!

5.2 Exam Problem 2: Initial ARP Request & Switch Self-Learning Workflow

Slide Exam Question 2 (Referencing Image 3): Host A (IP: 192.168.1.10, MAC: AA-AA-AA-AA-AA-AA), Host B (IP: 192.168.1.12, MAC: BB-BB-BB-BB-BB-BB) connected to Switch S1 at ports F1 and F2 respectively. Switch S1 and Switch S2 are connected via port F3. Host C (IP: 192.168.1.20, MAC: CC-CC-CC-CC-CC-CC) is connected to S2 at port F4. Initially, all ARP and MAC tables are empty. Host A wants to send a message to Host C.

- What kind of ARP packet (mention source and destination MAC address) will Host A generate, and how will the switches process it?
- Switches are termed "plug-and-play". Briefly explain the significance.

I. ARP Packet Generation and Processing:

- **Packet Generated by Host A:** Host A generates a **Broadcast ARP Request** packet because it knows Host C's IP (192.168.1.20) but does not know Host C's MAC address.
 - **Source MAC:** AA-AA-AA-AA-AA-AA (Host A)
 - **Destination MAC:** FF-FF-FF-FF-FF-FF (Broadcast)
 - **Source IP:** 192.168.1.10 | **Target IP:** 192.168.1.20
- **Processing by Switch S1:**
 - Frame enters S1 on port **F1**. S1 inspects Source MAC (AA-AA-AA-AA-AA-AA) and updates its MAC table: AA-AA-AA-AA-AA-AA → **F1**.
 - S1 inspects Destination MAC (FF-FF-FF-FF-FF-FF). Since it is a broadcast address, S1 **floods** the frame out all active ports except ingress port F1 (i.e., out port **F2** to Host B and port **F3** to Switch S2).
 - Host B receives the frame on F2, checks target IP (192.168.1.20), sees it does not match its own IP (192.168.1.12), and silently drops it.
- **Processing by Switch S2:**
 - Frame enters S2 on port **F3**. S2 inspects Source MAC (AA-AA-AA-AA-AA-AA) and updates its MAC table: AA-AA-AA-AA-AA-AA → **F3**.
 - S2 inspects Destination MAC (FF-FF-FF-FF-FF-FF) and **floods** the frame out all active ports except F3 (i.e., out port **F4** to Host C).
 - Host C receives the frame on F4, matches target IP (192.168.1.20), updates its local ARP table with Host A's MAC, and sends a **Unicast ARP Reply** back to Host A.

- Significance of "Plug-and-Play":** "Plug-and-play" signifies that switches do not require any manual network setup, administrative configuration, or pre-programmed topology maps. When plugged into a network, switches automatically construct their forwarding tables dynamically in real-time simply by inspecting the source MAC addresses of incoming traffic.

5.3 Exam Problem 3: Cross-Subnet ARP Analysis Across Routers

Slide Exam Question 3 (Referencing Image 8): Referring to the topology below: Hosts A, B, C connected to SW1. SW1 connected to Router R1 (interface fa0/1). R1 connected via s0/1 to Router R2 (s0/1). R2 (interface fa0/1) connected to SW2, which connects Hosts M and O.

1. State if host A will ever find the MAC address of host O in its ARP table.
2. Host A sends an ARP request for host C's MAC address. State which devices will receive that ARP request and what they will do with that ARP request.

1. **Will Host A ever find Host O's MAC address in its ARP table? No.** Host A and Host O reside on separate IP subnets separated by Layer 3 routers (R1 and R2). Routers do not forward Layer 2 broadcast ARP requests across subnet boundaries. When Host A communicates with Host O, Host A only resolves and stores the MAC address of its local ****Default Gateway (Router R1's fa0/1 interface)**** in its ARP table. Host A never learns Host O's MAC address.
2. **Host A sends ARP request for Host C's MAC address:** Since Host C is on the same local subnet, Host A broadcasts an ARP request frame (FF-FF-FF-FF-FF-FF). Switch SW1 receives the frame and floods it out all active ports.
 - **Devices Receiving the ARP Request:** (1) Host B, (2) Host C, and (3) Router R1 (interface fa0/1).
 - **Action Taken by Each Device:**
 - (a) **Host B:** Receives the broadcast frame, checks the target IP inside the payload, realizes it does not match its own IP, and ****silently drops/discards**** the packet.
 - (b) **Router R1 (fa0/1):** Receives the broadcast frame, inspects the target IP, sees it does not match its interface IP, and ****drops the packet****. Router R1 strictly ****does NOT forward**** the Layer 2 broadcast over serial link s0/1.
 - (c) **Host C:** Receives the broadcast frame, recognizes its own IP address as the target, updates its ARP cache with Host A's IP-to-MAC mapping, and generates a ****Unicast ARP Reply**** containing its MAC address back to Host A through SW1.

5.4 Exam Problem 4: Multi-Switch Self-Learning Simulation

Slide Exam Question 4 (Referencing Images 5 & 6): *Topology: S1, S2, S3, S4 interconnected in a line/tree structure. Ports: S1(F0→PC1, F1→PC2, F2→S2-F0); S2(F0→S1-F2, F1→PC3, F2→S3-F0); S3(F0→S2-F2, F1→PC4, F2→PC5, F3→S4-F0); S4(F0→S3-F3, F1→PC6).*

Suppose initial MAC tables of all switches are empty. PC2 sends two frames: one to PC4 and the other to PC6. How does switch S1 forward these two frames? Consider that PC4 does NOT reply, but PC6 DOES reply. Illustrate the current state of the MAC Address tables of switches S3 and S4. [Mention device name and interface]

Step-by-Step Simulation & Analysis:

1. **Frame Transmission from PC2:** Both Frame 1 (destined for PC4) and Frame 2 (destined for PC6) enter Switch S1 on interface **F1**.
2. **Action by Switch S1:**
 - S1 inspects ingress interface F1 and learns: $\text{MAC}_{\text{PC2}} \rightarrow \mathbf{F1}$.
 - For Frame 1 (to PC4): Destination MAC_{PC4} is unknown. S1 **floods** Frame 1 out ports **F0** (PC1) and **F2** (to S2).
 - For Frame 2 (to PC6): Destination MAC_{PC6} is unknown. S1 **floods** Frame 2 out ports **F0** (PC1) and **F2** (to S2).
3. **Propagation through S2 and S3:**
 - Switch S2 receives flooded frames on F0, learns $\text{MAC}_{\text{PC2}} \rightarrow \mathbf{F0}$, and floods out F1 (PC3) and F2 (S3).
 - Switch S3 receives flooded frames on F0, learns $\text{MAC}_{\text{PC2}} \rightarrow \mathbf{F0}$, and floods out F1 (PC4), F2 (PC5), and F3 (S4).
 - PC4 receives Frame 1 but **does NOT reply**. Therefore, no switch ever learns MAC_{PC4} .
4. **Propagation through S4 and PC6 Reply:**
 - Switch S4 receives Frame 2 on F0, learns $\text{MAC}_{\text{PC2}} \rightarrow \mathbf{F0}$, and forwards out F1 (PC6).
 - **PC6 Replies:** PC6 transmits a Unicast reply frame back to PC2.
 - The reply enters S4 on interface **F1**. S4 learns $\text{MAC}_{\text{PC6}} \rightarrow \mathbf{F1}$. S4 looks up MAC_{PC2} , finds it mapped to F0, and unicasts the reply out F0 to S3.
 - The reply enters S3 on interface **F3**. S3 learns $\text{MAC}_{\text{PC6}} \rightarrow \mathbf{F3}$. S3 looks up MAC_{PC2} , finds it mapped to F0, and unicasts the reply out F0 to S2.

5.4.1 Final MAC Address Tables for Switches S3 and S4

Switch S3 MAC Address Table:

Device	MAC Address	Interface Port
PC2		F0
PC6		F3

(Note: PC4 is NOT in S3's table because PC4 never sent a reply frame).

Switch S4 MAC Address Table:

Device MAC Address	Interface Port
PC2	F0
PC6	F1

5.5 Exam Problem 5: Unicasting vs. Flooding with Partial MAC Tables

Slide Exam Question 5 (Referencing Image 9): Referring to the figure below: Host A (00.90.21.D0.DD.48) connected to SW1-Fa0/1; Host B (00.04.9A.10.C6.78) connected to SW1-Fa0/2; SW1-Fa0/3 connected to SW2-Fa0/1. Host C (00.01.C9.59.7A.C2) connected to SW2-Fa0/2; Host D (00.0C.85.75.64.90) connected to SW2-Fa0/3.

Initially, the MAC address table of switch SW1 contains information for Host D & Host A, while SW2 is EMPTY. Now, Host A sends a packet to Host D. Explain what the two switches do with this packet in terms of unicasting and flooding. Show the updated MAC address tables of both switches.

Step-by-Step Solution:

1. Action by Switch SW1 (Unicasting):

- Host A sends a frame with Source MAC = Host A (00.90.21.D0.DD.48) and Destination MAC = Host D (00.0C.85.75.64.90).
- Frame arrives at SW1 on port **Fa0/1**. SW1 verifies/refreshes Host A on Fa0/1.
- SW1 looks up Destination MAC (00.0C.85.75.64.90) in its table. Since SW1 already contains Host D mapped to **Fa0/3**, SW1 performs **UNICASTING**.
- SW1 selectively forwards the frame **strictly out interface Fa0/3** towards SW2. SW1 does **NOT** flood the frame to Host B on Fa0/2.

2. Action by Switch SW2 (Flooding):

- Frame arrives at SW2 on ingress port **Fa0/1**.
- SW2 inspects Source MAC (00.90.21.D0.DD.48) and **learns** Host A on port Fa0/1.
- SW2 looks up Destination MAC (00.0C.85.75.64.90) in its table. Because SW2's table was initially empty, Host D is **UNKNOWN** to SW2.
- Therefore, SW2 performs **FLOODING** and transmits the frame out all active ports except ingress port Fa0/1 (i.e., out **Fa0/2** to Host C and **Fa0/3** to Host D).
- Host D receives the packet intended for it. Host C receives it but drops it at Layer 2.

5.5.1 Updated MAC Address Tables for SW1 and SW2

Table 5: Updated MAC Address Tables after Host A sends packet to Host D

Switch SW1 MAC Table		Switch SW2 MAC Table	
MAC Address	Interface	MAC Address	Interface
Host A (00.90.21.D0.DD.48)	Fa0/1	Host A (00.90.21.D0.DD.48)	Fa0/1
Host D (00.0C.85.75.64.90)	Fa0/3	–	–

Important Note on SW2's Table: Host D is **NOT** yet learned by SW2 at this stage because Host D has not transmitted a reply frame back through SW2. Once Host D sends a reply frame back to Host A, SW2 will learn Host D on interface Fa0/3.